AP Biology — Unit 2 Practice Test

Cell Structure and Function · Membrane Transport

Directions: Each of the questions below is followed by four suggested answers. Select the one that is best in each case. Mark your answer on the answer sheet.

1. Which organelle is the main site of ATP production in a eukaryotic cell?

   (A) Ribosome
   (B) Mitochondrion
   (C) Golgi apparatus
   (D) Lysosome

2. A cell is placed in a solution and swells. Compared with the cell, the solution is

   (A) hypertonic
   (B) isotonic
   (C) hypotonic
   (D) saturated

3. Which of the following moves across the membrane without a transport protein?

   (A) Glucose
   (B) Sodium ions
   (C) Oxygen
   (D) Amino acids

4. Why do small cells exchange materials with their surroundings more efficiently than large cells?

   (A) They have more organelles
   (B) Their surface area to volume ratio is higher
   (C) Their membranes are thinner
   (D) They contain less water

5. Which structure is found in plant cells but not in animal cells?

   (A) Mitochondrion
   (B) Plasma membrane
   (C) Cell wall
   (D) Nucleus

6. The sodium-potassium pump is an example of

   (A) simple diffusion
   (B) facilitated diffusion
   (C) osmosis
   (D) active transport

7. Proteins destined for secretion are first assembled on ribosomes attached to the

   (A) smooth ER
   (B) rough ER

(C) Golgi apparatus
(D) nuclear envelope

8. Water potential of pure water at atmospheric pressure is

(A) positive
(B) negative
(C) zero
(D) undefined

9. Which best describes the fluid mosaic model?

(A) A rigid bilayer of proteins
(B) A flexible phospholipid bilayer with embedded proteins
(C) A single layer of lipids
(D) A wall of cellulose fibres

10. Aquaporins increase the rate of

(A) ion transport
(B) protein synthesis
(C) water movement
(D) ATP hydrolysis

11. Which organelle breaks down worn-out cell parts?

(A) Lysosome
(B) Peroxisome
(C) Vacuole
(D) Centrosome

12. Endosymbiotic theory is supported by the observation that mitochondria

(A) have a single membrane
(B) lack ribosomes
(C) contain their own circular DNA
(D) are found only in plants

Answer Key

1. B
2. C
3. C
4. B
5. C
6. D
7. B
8. C
9. B
10. C
11. A
12. C